TEXAS A&M UNIVERSITY

Sets

CSCE 222: Discrete Structures for Computing

Outline

- Sets
- Sets operations

Sets

What is a set?

Applications

Combinations: unordered collections of objects

Relations: sets of ordered pairs represent relationships between objects

Graphs: sets of ordered or unordered pairs also represent relationships

Functions

The Language of Sets

Let S be the set of all integers between 1 and 5, inclusive

$S = \{1,2,3,4,5\}$

We want to determine if x is an element of set S:

$x \in S$?

Let $x = 3$; is $x \in S$? Yes

Now let $x = 10$; is $x \in S$? No, so we denote this as $x \notin S$

The Language of Sets

Let set T = {5, 4, 3, 2, 1} and S = {1, 2, 3, 4, 5}

is set T = set S?

Do sets T and S contain the same elements?

Does {2} = 2?

{2} is a set with the element 2, while 2 is the number 2

Common Sets

C = set of complex numbers

R = set of real numbers

R^+ = set of positive real numbers

Q = set of rational numbers

Z = integers = {…,–3,–2,–1,0,1,2,3,…}

Z^+ = positive integers = {1,2,3,…}

N = natural numbers = {0,1,2,3….}

Common Sets

Recall the definition of rational numbers: can be written as a fraction and has a terminating or repeating decimal

Example: 5/2 = 2.5, 2.5 is a rational number

A number that is not rational is called irrational: non-terminating decimal.

π and SQRT(2) are irrational numbers, so they are not elements of set Q.

Set-Roster Notation

Set S = {1,2,3,4,5}

For a small set, we can write all the elements in the set. This is the set-roster notation (roster of elements).

For a large set, we can write A = {1,2,3,….100}.

That is all integers from 1 to 100, inclusive.

For some sets (e.g. real numbers), the roster notation is impractical.

Set-Builder Notation

We define the set of all elements x in set S such that P(x) is true as: $\{x \in S \mid P(x)\}$

1. $x \in S$ means "elements x in set S"
2. $\mid$ means "such that"
3. P(x) means that property P(x) is satisfied (true)

e.g. $\{x \in Z \mid 2 < x \leq 5\} = \{3, 4, 5\}$

Set Equality

When are two sets equal?

Set Equality

B = {1,2,3,4,5}

D = {4,1,3,5,2}

E = {1,2,3,4}

Is B equal to D?

Is B equal to E?

Set Equality – Formal Definition

Given sets A and B, A equals B (A = B) if and only if:

Every element of A is in B, and

Every element of B is in A

$$A = B \Leftrightarrow (A \subseteq B) \wedge (B \subseteq A)$$

$$\forall x\, [x \in A \leftrightarrow x \in B]$$

The Empty Set { } or ∅

∅ or { } is the empty set, the set that has no elements.

The empty set has a cardinality of 0.

Note: a set with one element is called the singleton set.

Do not get confused!

∅ or { } means the empty set.

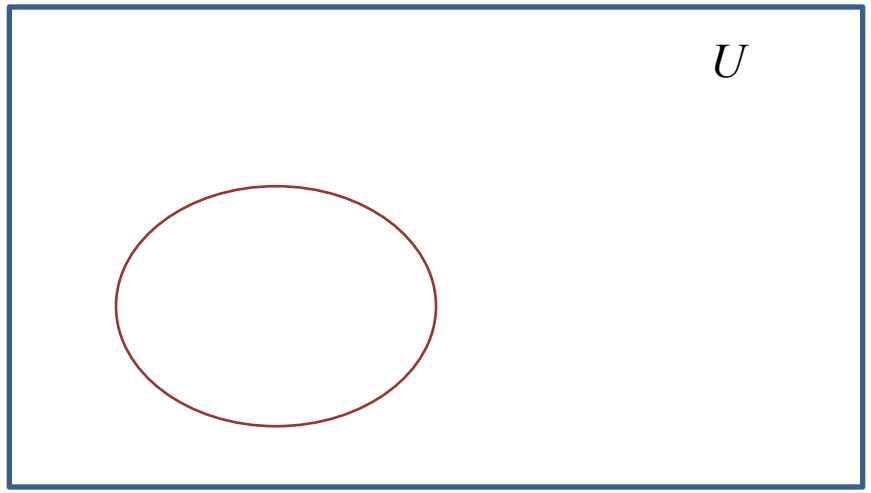
{∅} is a singleton set containing the empty set.

Venn Diagrams

Venn diagrams are visual representations of sets.

The Universal set (all objects under consideration) is a rectangle.

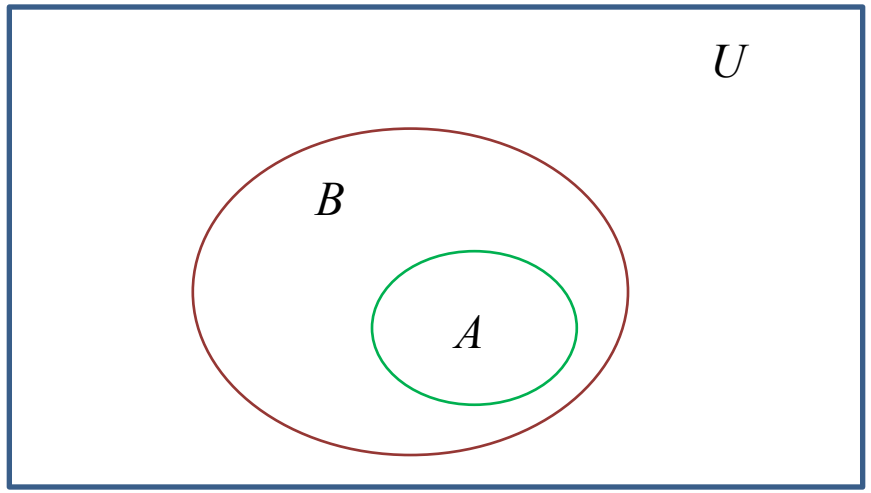
Particular sets are circles inside the rectangle.



Subsets

Set A is a subset of set B if every element of A is in B.

Notation: $A \subseteq B$



Subsets

Assume we have the following sets:

$A = Z$ (set of integers)
$B = \{1,2,3,4,5\}$ (set-roster notation)
$C = \{x \in Z \mid 2 < x < 5\}$ (set-builder notation)

All the elements of set B are in set A; B is a subset of A: $B \subseteq A$

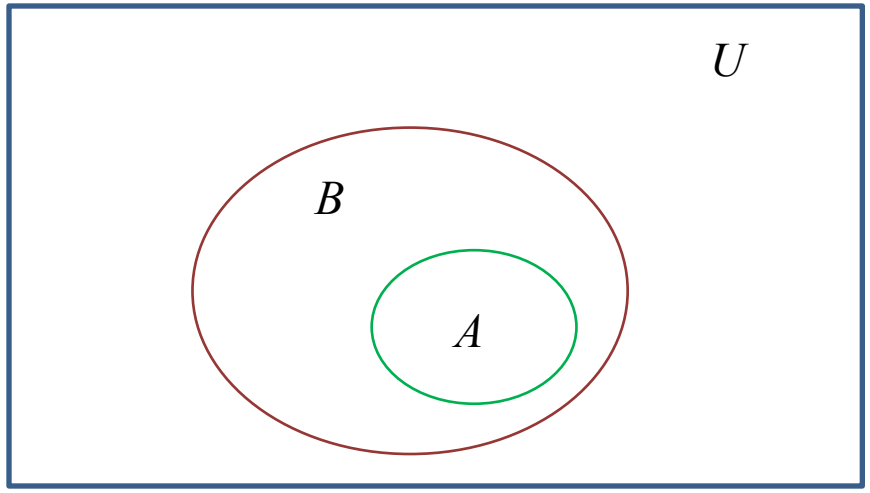
$B \subseteq A$ means that for all x, if $x \in B$, then $x \in A$

What about set C ?

Proper subsets

Set A is a proper subset of B if A is a subset of B but is not equal to B. Notation: $A \subset B$

Every element of A is in B and there is at least one element of B that is not in A.



Subset vs. Proper Subset

Assume we have the following sets:
B = {1,2,3,4,5}
D = {4,1,3,5,2}
E = {2,4,1,3}

Note that D = B

Recall the definition of a subset: $\forall x$, if $x \in D$, then $x \in B$.

We can state that D is a subset of B.

We can also state that B is a subset of itself.

However, E is a proper subset of B since there is at least one element in B that is not in E (5).

Power set

The power set of a set S is the set of all subsets of S.
Notation: P(S)

e.g.

- The power set of $\varnothing$ (cardinality 0) is $\{\varnothing\}$ (cardinality 1)
- The power set of {0} (cardinality 1) is $\{\varnothing, \{0\}\}$ (cardinality 2)
- The power set of {0,1} (cardinality 2) is $\{\varnothing,\{0\},\{1\},\{0,1\}\}$ (cardinality 4)

Note: if $|S| = n$, then $|P(S)| = 2^n$

Operations on Sets

Earlier, we discussed operations on statement variables ($\vee$, $\wedge$,~)

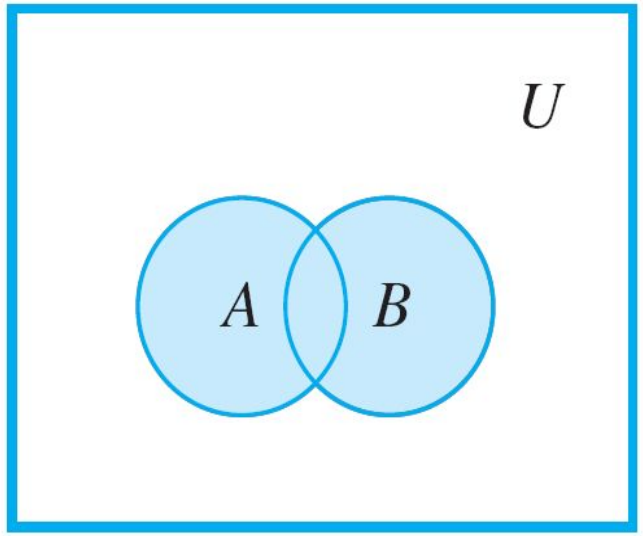
There are similar operations that we perform on sets:

- Union
- Intersection
- Complement
- Difference

Sets Union

The union of sets A and B is the set of all elements that are in at least one of A or B

Denoted as $A \cup B$

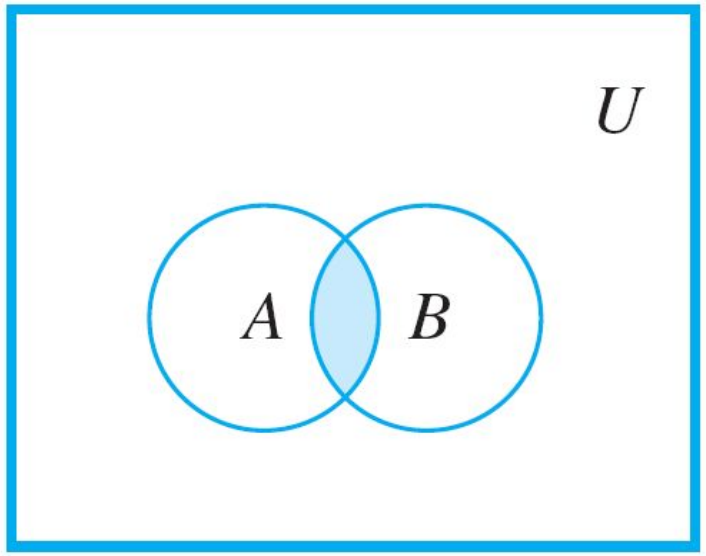


Shaded region represents $A \cup B$.

Sets Intersection

The intersection of sets A and B is the set of all elements that common to both A or B

Denoted as $A \cap B$

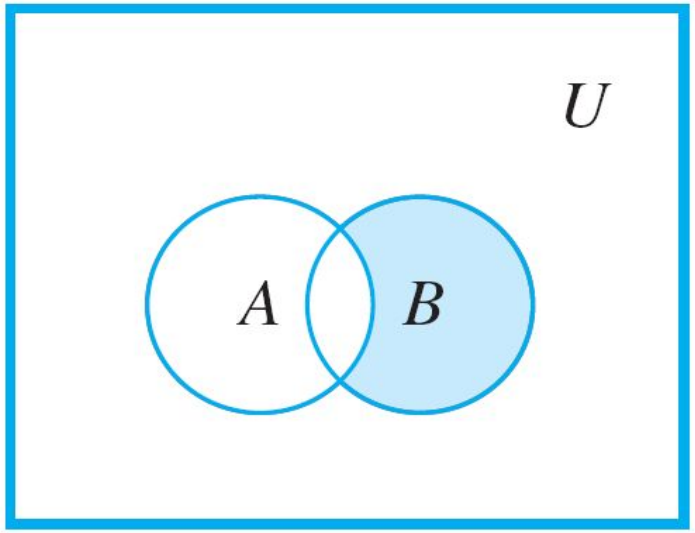


Shaded region represents $A \cap B$.

Sets Difference

The difference B minus A is the set of all elements that are in B and not A

Denoted as $B - A$

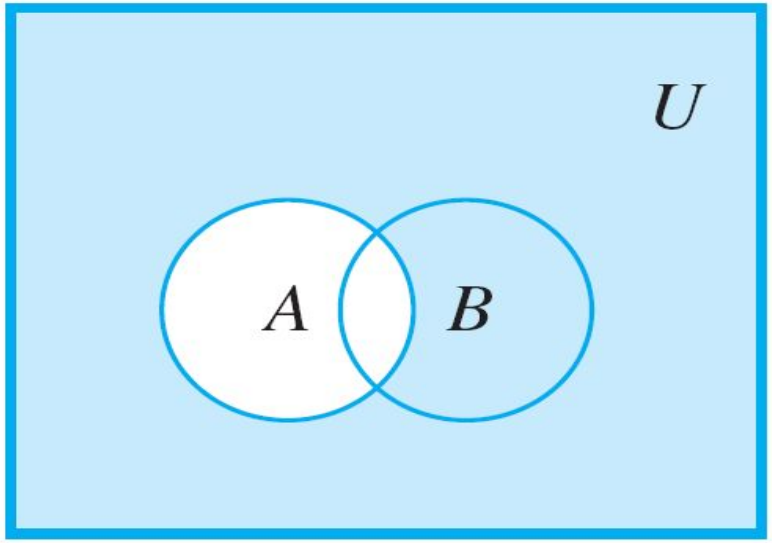


Shaded region represents $B - A$.

Set Complement

The complement of A is the set of all elements in U that are not in A

Denoted as A^c



Shaded region represents A^c.

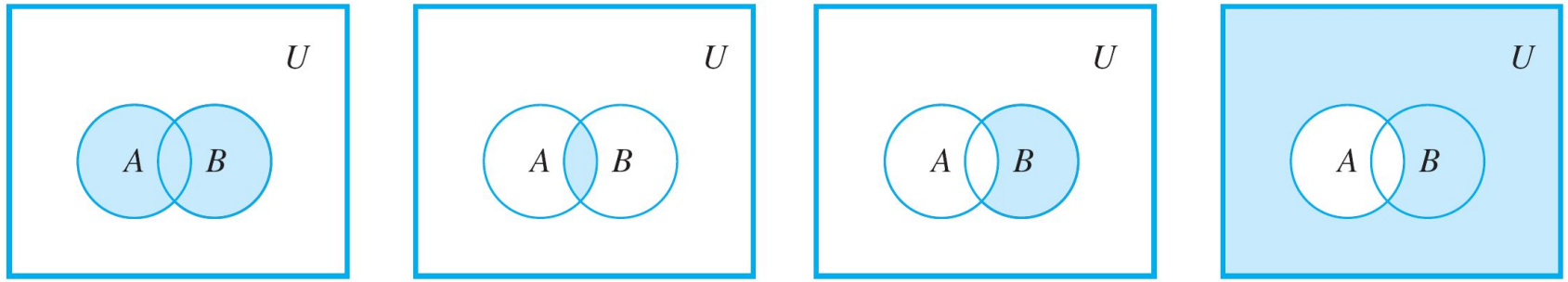
Formal Definitions of Set Operators

$$A \cup B = \{x \in U \mid x \in A \vee x \in B\}$$

$$A \cap B = \{x \in U \mid x \in A \wedge x \in B\}$$

$$B - A = \{x \in U \mid x \in B \wedge x \notin A\}$$

$$A^c = \{x \in U \mid x \notin A\}$$



Activity

Let the universal set $U = \{1,2,3,4,5,6,7\}$

Set $A = \{1,3,5,7\}$, Set $B = \{4,5,6,7\}$

$A \cup B = ?$

$A \cap B = ?$

$B - A = ?$

$A^c = ?$

Set Identities

Set identities are similar to the logical equivalences we learned in Ch.1. Here are De Morgan's Laws for sets:

$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

We also have identity, domination, idempotent, double complementation, commutative, associative, absorption, complement.

Read Table 1 in Section 2.2, page 130

Cartesian Product

The Cartesian product of sets A and B is the set of all ordered pairs (a, b) where a is in set A and b is in set B

Or: $$A \times B = \{(a,b) \mid a \in A \text{ and } b \in B\}$$

Given A = {3,5,7}, B = {2,4}

A×B = {(3,2), (3,4), (5,2), (5,4), (7,2), (7,4)}

B×A = {(2,3), (2,5), (2,7), (4,3), (4,5), (4,7)}

B×B = {(2,2), (2,4), (4,2), (4,4)}

If |A| = n and |B| = m, What is |A × B|?

Relations

Given two sets A and B:

A relation R from A to B is a subset of A×B.

$R \subseteq A \times B$

Given an ordered pair (x,y) in A×B:

x is related to y by R iff (x,y) is in R

A is the domain of R and B is the co-domain of R

Activity

$A = \{3,5,7\}$, $B = \{2,4\}$

Given any $(x, y) \in A \times B$

Define a relation R from A to B as follows:

1. $(x, y) \in R$ means $x < y$
2. $(x, y) \in R$ means $y = x - 1$

Arrow Diagram of a Relation

We can also represent relations graphically using an arrow diagram:

The diagram represents sets as regions, with elements of sets as points in regions.

For each x in the domain region and y in the co-domain region, draw an arrow from x to y iff $(x,y) \in R$

Example: Arrow Diagrams of Relations

Let $A = \{1, 2, 3\}$ and $B = \{1, 3, 5\}$

For all $(x, y) \in A \times B$

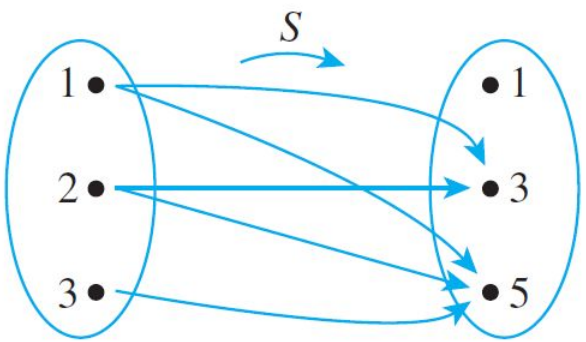
Let's define relations S and T from A to B as follows:

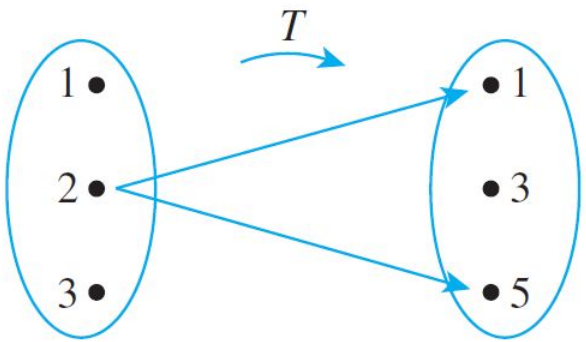
$(x, y) \in S$ means that $x < y$

$T = \{(2, 1), (2, 5)\}$

Draw arrow diagrams for S and T.

Example: Solution





It is possible to have arrows from the same element of *A* pointing to different directions.

It is possible to have an element of *A* that does not have an arrow coming out of it.